07_bayesian_model

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1 STATISTICAL LEARNING FINAL PROJECT

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[36]:
```
import pandas as pd
import numpy as np
import pymc as pm
import arviz as az
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

df = pd.read_csv('/content/drive/MyDrive/feature_engineered_real_estate.csv')
df.head()
```

[36]:
```
   TotalFinishedArea  LivingUnits  TotalAppraisedValue  SaleYear  SaleMonth  \
0             1092.0          1.0               187200      2024          7
1              936.0          1.0               192300      2022          4
2             1111.0          1.0               148400      2023         12
3             1684.0          1.0               197900      2021          1
4              900.0          1.0               170700      2021          9

   PriceRatio  AssrLandUse_APT FOUR  AssrLandUse_CONDOMINIMUM  \
0    1.383547                 False                     False
1    1.081643                 False                     False
2    0.842318                 False                     False
3    0.894391                 False                     False
4    1.101347                 False                     False

   AssrLandUse_MULTI DWLG  AssrLandUse_ONE FAMILY  AssrLandUse_THREE FAMILY  \
0                   False                    True                     False
1                   False                    True                     False
2                   False                    True                     False
3                   False                    True                     False
4                   False                    True                     False

   AssrLandUse_TWO FAMILY
```

```
0                    False
1                    False
2                    False
3                    False
4                    False
```

2 Step 1

[37]:
```python
# Check data
print(df.shape)
print(df.columns)
print(df.isnull().sum())
```

```
(3137, 12)
Index(['TotalFinishedArea', 'LivingUnits', 'TotalAppraisedValue', 'SaleYear',
       'SaleMonth', 'PriceRatio', 'AssrLandUse_APT FOUR',
       'AssrLandUse_CONDOMINIMUM', 'AssrLandUse_MULTI DWLG',
       'AssrLandUse_ONE FAMILY', 'AssrLandUse_THREE FAMILY',
       'AssrLandUse_TWO FAMILY'],
      dtype='object')
TotalFinishedArea           0
LivingUnits                 0
TotalAppraisedValue         0
SaleYear                    0
SaleMonth                   0
PriceRatio                  0
AssrLandUse_APT FOUR        0
AssrLandUse_CONDOMINIMUM    0
AssrLandUse_MULTI DWLG      0
AssrLandUse_ONE FAMILY      0
AssrLandUse_THREE FAMILY    0
AssrLandUse_TWO FAMILY      0
dtype: int64
```

[38]:
```python
# Define response and predictors
y_col = "PriceRatio"

x_cols = [
    "TotalFinishedArea",
    "LivingUnits",
    "SaleYear",
    "SaleMonth",
    "AssrLandUse_APT FOUR",
    "AssrLandUse_CONDOMINIMUM",
    "AssrLandUse_MULTI DWLG",
    "AssrLandUse_THREE FAMILY",
```

```
    "AssrLandUse_TWO FAMILY"
]
```

[39]:
```
df = df[[y_col] + x_cols].copy()
```

[40]:
```
# Split into training and test sets
df_train, df_test = train_test_split(df, test_size=0.2, random_state=6243)

print(df_train.shape)
print(df_test.shape)
```

```
(2509, 10)
(628, 10)
```

[41]:
```
# Standardize only the continuous predictors
cont_cols = [
    "TotalFinishedArea",
    "LivingUnits",
    "SaleYear",
    "SaleMonth"
]

scaler = StandardScaler()

df_train = df_train.copy()
df_test = df_test.copy()

df_train[cont_cols] = scaler.fit_transform(df_train[cont_cols])
df_test[cont_cols] = scaler.transform(df_test[cont_cols])

df_train.head()
```

[41]:
```
      PriceRatio  TotalFinishedArea  LivingUnits  SaleYear  SaleMonth  \
625     0.788397           0.987019     0.338207 -0.012819   0.670056
2623    1.022044           1.197209     1.494284 -0.882083   0.956213
211     1.122914          -1.160396    -0.817871 -0.012819   0.670056
2552    0.669291           0.374654     0.338207 -0.882083  -0.474571
1687    0.988372          -1.122330    -0.817871 -0.012819  -0.188414

      AssrLandUse_APT FOUR  AssrLandUse_CONDOMINIMUM  AssrLandUse_MULTI DWLG  \
625                  False                     False                   False
2623                 False                     False                   False
211                  False                     False                   False
2552                 False                     False                   False
1687                 False                      True                   False

      AssrLandUse_THREE FAMILY  AssrLandUse_TWO FAMILY
```

```
625                     False                     True
2623                     True                    False
211                     False                    False
2552                    False                     True
1687                    False                    False
```

3 Step 2

[42]:
```
with pm.Model() as bayes_model:
    # Priors
    intercept = pm.StudentT("Intercept", nu=4, mu=0, sigma=10)

    w_TotalFinishedArea   = pm.Normal("w_TotalFinishedArea", mu=0, sigma=10)
    w_LivingUnits         = pm.Normal("w_LivingUnits", mu=0, sigma=10)
    w_SaleYear            = pm.Normal("w_SaleYear", mu=0, sigma=10)
    w_SaleMonth           = pm.Normal("w_SaleMonth", mu=0, sigma=10)

    w_APT_FOUR            = pm.Normal("w_APT_FOUR", mu=0, sigma=10)
    w_CONDOMINIMUM        = pm.Normal("w_CONDOMINIMUM", mu=0, sigma=10)
    w_MULTI_DWLG          = pm.Normal("w_MULTI_DWLG", mu=0, sigma=10)
    w_THREE_FAMILY        = pm.Normal("w_THREE_FAMILY", mu=0, sigma=10)
    w_TWO_FAMILY          = pm.Normal("w_TWO_FAMILY", mu=0, sigma=10)

    sigma = pm.HalfNormal("sigma", sigma=1)

    # Linear predictor
    mu = (
        intercept
        + w_TotalFinishedArea   * df_train["TotalFinishedArea"]
        + w_LivingUnits         * df_train["LivingUnits"]
        + w_SaleYear            * df_train["SaleYear"]
        + w_SaleMonth           * df_train["SaleMonth"]
        + w_APT_FOUR            * df_train["AssrLandUse_APT FOUR"]
        + w_CONDOMINIMUM        * df_train["AssrLandUse_CONDOMINIMUM"]
        + w_MULTI_DWLG          * df_train["AssrLandUse_MULTI DWLG"]
        + w_THREE_FAMILY        * df_train["AssrLandUse_THREE FAMILY"]
        + w_TWO_FAMILY          * df_train["AssrLandUse_TWO FAMILY"]
    )

    # Likelihood
    y_obs = pm.Normal("PriceRatio", mu=mu, sigma=sigma,␣
↪observed=df_train[y_col])

    # Posterior sampling
    trace_bayes = pm.sample(
        draws=1000,
```

```
        tune=1000,
        chains=2,
        random_seed=6243,
        target_accept=0.9,
        return_inferencedata=True
    )
```

```
Output()
```

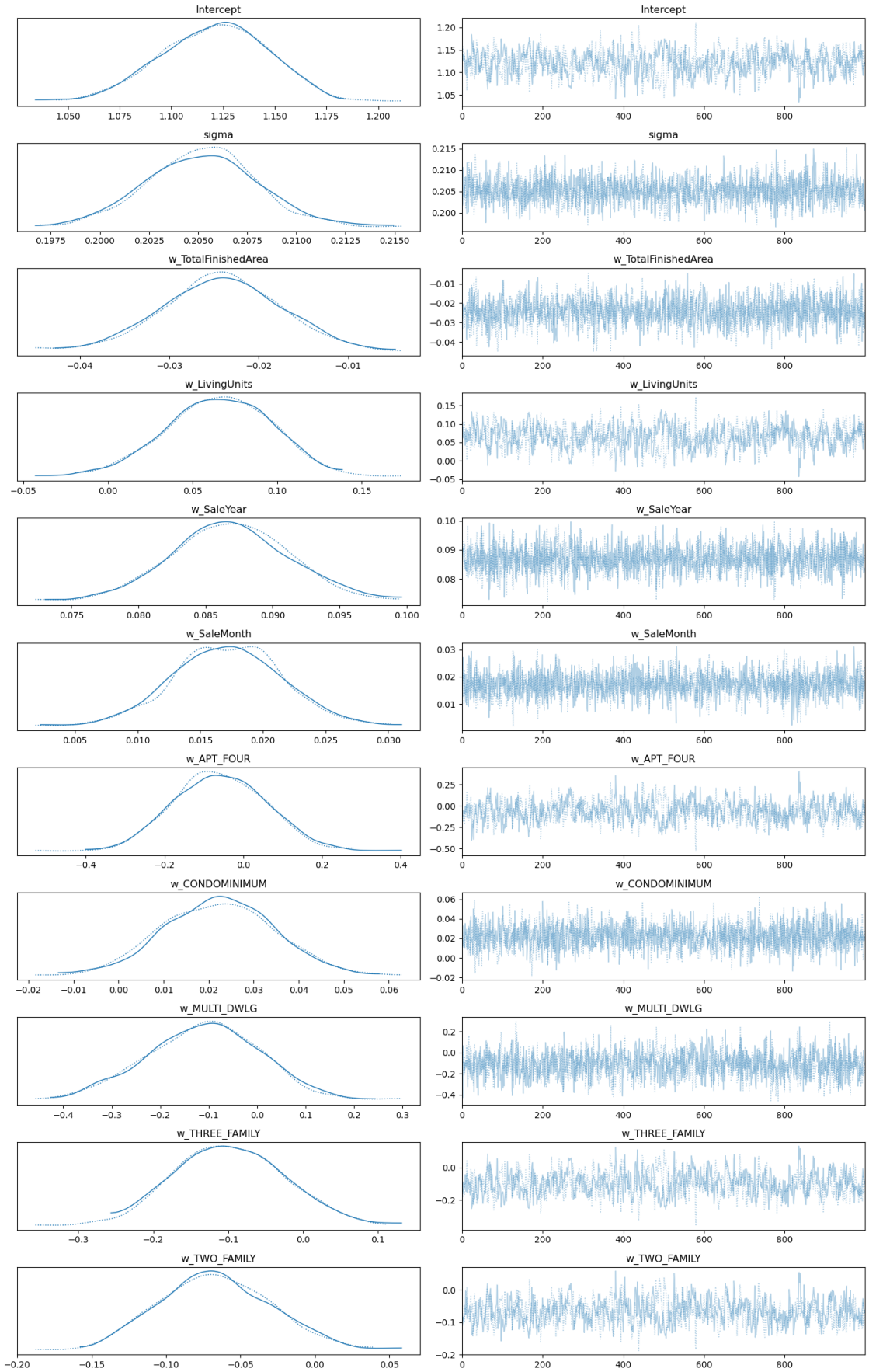
4 Step 3

[43]:
```
# Posterior summaries

# Summary table
summary = az.summary(trace_bayes, round_to=4)
print(summary)
```

```
                       mean      sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  \
Intercept            1.1208  0.0259  1.0733   1.1683     0.0012   0.0007
sigma                0.2052  0.0028  0.1997   0.2104     0.0001   0.0001
w_APT_FOUR          -0.0586  0.1165 -0.2637   0.1513     0.0053   0.0030
w_CONDOMINIMUM       0.0222  0.0123 -0.0001   0.0460     0.0003   0.0002
w_LivingUnits        0.0655  0.0307  0.0066   0.1199     0.0014   0.0008
w_MULTI_DWLG        -0.1087  0.1186 -0.3375   0.1021     0.0036   0.0026
w_SaleMonth          0.0172  0.0044  0.0087   0.0252     0.0001   0.0001
w_SaleYear           0.0868  0.0044  0.0789   0.0955     0.0001   0.0001
w_THREE_FAMILY      -0.0976  0.0723 -0.2282   0.0352     0.0034   0.0019
w_TWO_FAMILY        -0.0665  0.0375 -0.1348   0.0033     0.0018   0.0010
w_TotalFinishedArea -0.0241  0.0067 -0.0373  -0.0120     0.0002   0.0001

                      ess_bulk   ess_tail   r_hat
Intercept             462.3047   693.4344  1.0034
sigma                2022.3364  1666.2319  1.0019
w_APT_FOUR            480.0531   775.5453  1.0035
w_CONDOMINIMUM       2024.1803  1569.2713  1.0012
w_LivingUnits         459.0657   779.1961  1.0022
w_MULTI_DWLG         1091.3912  1334.5348  1.0008
w_SaleMonth          2269.1700  1289.2379  1.0003
w_SaleYear           2267.2030  1586.1813  1.0002
w_THREE_FAMILY        451.7577   769.2572  1.0035
w_TWO_FAMILY          454.8796   673.6675  1.0027
w_TotalFinishedArea  1505.9225  1548.7904  1.0062
```

[44]:
```
az.plot_trace(
    trace_bayes,
```

```
    var_names=[
        "Intercept", "sigma",
        "w_TotalFinishedArea", "w_LivingUnits",
        "w_SaleYear", "w_SaleMonth",
        "w_APT_FOUR", "w_CONDOMINIMUM", "w_MULTI_DWLG",
        "w_THREE_FAMILY", "w_TWO_FAMILY"
    ],
    figsize=(14, 22)
)
plt.tight_layout()
plt.show()
```


Intercept
sigma
w_TotalFinishedArea
w_LivingUnits
w_SaleYear
w_SaleMonth
w_APT_FOUR
w_CONDOMINIMUM
w_MULTI_DWLG
w_THREE_FAMILY
w_TWO_FAMILY

[ ]:
```
!apt-get update
!apt-get install -y texlive-xetex texlive-fonts-recommended␣
↪texlive-plain-generic
```

```
 0% [Working]              Hit:1 http://archive.ubuntu.com/ubuntu jammy
InRelease
 0% [Connecting to security.ubuntu.com (91.189.91.82)] [Connecting to cloud.r-pr
Hit:2 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
 Hit:3 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
 0% [Connecting to security.ubuntu.com (91.189.91.82)] [Connected to cloud.r-pro
Hit:4 https://cli.github.com/packages stable InRelease
 0% [Connecting to security.ubuntu.com (91.189.91.82)] [Connected to cloud.r-pro
Hit:5 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/ InRelease
 0% [Waiting for headers] [Connecting to r2u.stat.illinois.edu (192.17.190.167)]
Hit:6 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy InRelease
 0% [Waiting for headers] [Connecting to r2u.stat.illinois.edu (192.17.190.167)]
0% [Waiting for headers] [Connecting to r2u.stat.illinois.edu (192.17.190.167)]
Hit:7 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:8 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy InRelease
Hit:9 https://r2u.stat.illinois.edu/ubuntu jammy InRelease
Reading package lists… Done
W: Skipping acquire of configured file 'main/source/Sources' as repository
'https://r2u.stat.illinois.edu/ubuntu jammy InRelease' does not seem to provide
it (sources.list entry misspelt?)
```

[ ]:
```
!jupyter nbconvert --to pdf '/content/drive/MyDrive/Colab Notebooks/
↪07_bayesian_model.ipynb'
```